Institute of Technology of Cambodia

Department of Information and Communication Engineering

Assignment: Lab04

Subject: **Operating System**

Lecturer Course/TP: **Heng Rathpisey**

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Year: **I4**

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1. Create 3 folders in root directory: Engineering, Sale, IS

```
sokmeak@Labtop-SokmeakSR:/$ sudo mkdir Engineering Sale IS;
sokmeak@Labtop-SokmeakSR:/$ ls
Engineering  bin                dev   init                lib64       mnt   root  sbin.usr-is-merged  sys   usr
IS           bin.usr-is-merged  etc   lib                 lost+found  opt   run   snap                test  var
Sale         boot               home  lib.usr-is-merged   media       proc  sbin  srv                 tmp
```

2. Create 3 groups: Engineering, Sale, IS

```
sokmeak@Labtop-SokmeakSR:/$ sudo groupadd Engineering;
sokmeak@Labtop-SokmeakSR:/$ sudo groupadd Sale;
sokmeak@Labtop-SokmeakSR:/$ sudo groupadd IS;
sokmeak@Labtop-SokmeakSR:/$ cat /etc/group
```

```
Engineering:x:1002:
Sale:x:1003:
IS:x:1004:
```

3. Create admin user in each department: admineng, adminsale, adminIS.

```
sokmeak@Labtop-SokmeakSR:/$ sudo useradd -m -s /bin/bash admineng;
useradd: warning: the home directory /home/admineng already exists.
useradd: Not copying any file from skel directory into it.
```

```
sokmeak@Labtop-SokmeakSR:/$ sudo useradd -m -s /bin/bash adminIS;
useradd: warning: the home directory /home/adminIS already exists.
useradd: Not copying any file from skel directory into it.
sokmeak@Labtop-SokmeakSR:/$ sudo useradd -m -s /bin/bash adminsale;
useradd: warning: the home directory /home/adminsale already exists.
useradd: Not copying any file from skel directory into it.
sokmeak@Labtop-SokmeakSR:/$ cat /etc/passwd;
```

```
admineng:x:1002:1005::/home/admineng:/bin/bash
adminIS:x:1003:1006::/home/adminIS:/bin/bash
adminsale:x:1004:1007::/home/adminsale:/bin/bash
```

a. Add these admin user to sudo group as secondary group.

```
sokmeak@Labtop-SokmeakSR:/$ sudo usermod -G sudo admineng; sudo usermod -G sudo adminIS; sudo usermod -G sudo adminsale;
```

b. Add these admin user to each department as primary group.

```
sokmeak@Labtop-SokmeakSR:/$ sudo usermod -g Engineering admineng; sudo usermod -g IS adminIS; sudo usermod -g Sale adminsale;
```

```
sokmeak@Labtop-SokmeakSR:/$ groups admineng;
admineng : Engineering sudo
sokmeak@Labtop-SokmeakSR:/$ groups adminsale;
adminsale : Sale sudo
sokmeak@Labtop-SokmeakSR:/$ groups adminIS;
adminIS : IS sudo
sokmeak@Labtop-SokmeakSR:/$
```

4. Create 2 normal users in each departpartment and then add these users ot each department group as primary group

```
sokmeak@Labtop-SokmeakSR:/$ sudo usermod -g Engineering eng1; sudo usermod -g Engineering eng2;
sokmeak@Labtop-SokmeakSR:/$ sudo useradd -m -s /bin/bash IS1;
sokmeak@Labtop-SokmeakSR:/$ sudo useradd -m -s /bin/bash IS2;
sokmeak@Labtop-SokmeakSR:/$ sudo usermod -g IS IS1; sudo usermod -g IS IS2;
sokmeak@Labtop-SokmeakSR:/$ sudo useradd -m -s /bin/bash sale1;
sokmeak@Labtop-SokmeakSR:/$ sudo useradd -m -s /bin/bash sale2;
sokmeak@Labtop-SokmeakSR:/$ sudo usermod -g Sale sale1; sudo usermod -g Sale sale2;
sokmeak@Labtop-SokmeakSR:/$
```